GEO Satellite Debris Observation Scheduling System

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Overview

Developed a real-world scheduling system during internship at **U R Rao Satellite Centre (ISRO)** to optimize observation of GEO space debris using a ground-based optical telescope under strict physical and operational constraints.

Problem Statement

- Hundreds of GEO debris objects are visible each night
- Telescope can observe only **~166 objects/night (physical limit)**
- Constraints:
    - Fixed observation time (90s)
    - Slew/settling time (120s)
    - Night-only operation
    - Non-overlapping scheduling

Goal: **Maximize observation efficiency under real-world constraints**

Solution Approach

1. Greedy Scheduling (EDF)

- Implemented Earliest Deadline First strategy
- Achieved **near-maximum capacity (~165–166 observations/night)**
- Ensured:
    - Feasible schedules
    - High utilization of telescope time

2. Genetic Algorithm Optimization

- Designed GA for **global scheduling optimization**
- Chromosome: permutation of debris observation order
- Fitness based on:
    - Number of observations
    - Observation quality (peak elevation)

Result:

- Improved **observation quality (higher elevation)**
- Handled heavy overlap better than greedy

3. Multi-Night Scheduling Framework

- Extended system to **5-night scheduling pipeline**
- Removed already observed debris after each night
- Achieved **near-complete debris coverage over multiple nights**

Key Results

- Greedy: ~166 observations/night (near physical limit)
- GA: Fewer observations but **higher-quality data**

- Demonstrated real trade-off:
    - **Quantity vs Quality optimization**

Key Contributions

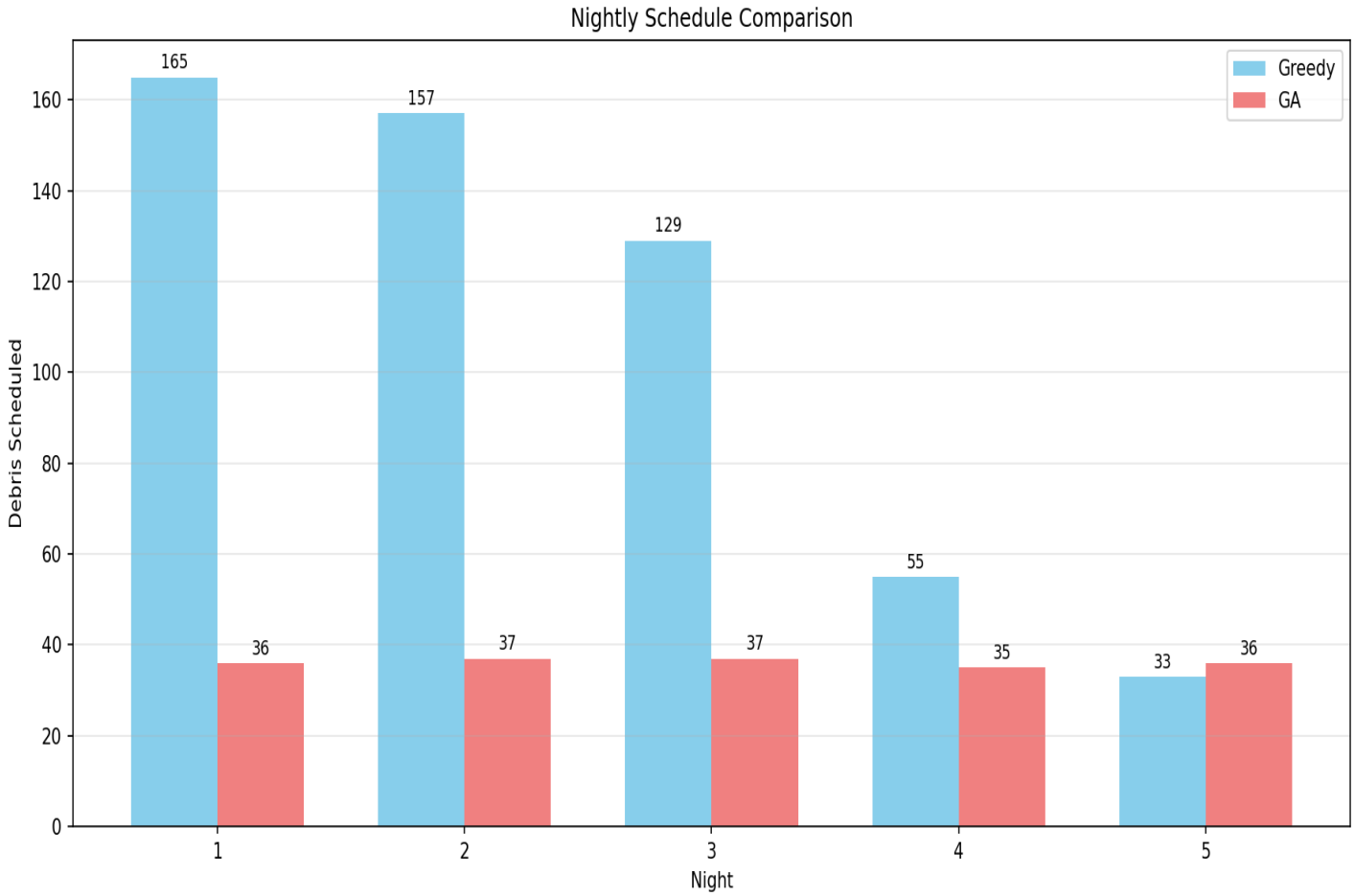
- Modeled problem as **constraint-based scheduling (not detection)**
- Built **end-to-end pipeline**:
    - Data preprocessing
    - Constraint modeling
    - Scheduling algorithms
    - Performance evaluation
- Reflected **real ISRO/NASA operational practices**

Tech Stack

Python • Optimization Algorithms • Genetic Algorithms • Data Processing • Scheduling Systems

Impact

- Simulates real **Space Situational Awareness (SSA)** workflows
- Applicable to:
    - Satellite scheduling
    - Telescope planning
    - Resource-constrained optimization systems



Greedy achieves near-maximum throughput (~166/night), while GA optimizes for observation quality, demonstrating a real-world trade-off between quantity and quality.